

Question 20

Not yet answered

Marked out of 2.00

Flag question

In Backpropagation learning algorithm, the change of the transfer function with respect to the change of the input to the activation function represents:

- ☒ a. Sensitivity
- ☐ b. Training Set
- ☐ c. Bias
- ☐ d. Actual Outputs
- ☐ e. Targets
- ☐ f. Forward Propagation
- ☐ g. Weight

[Clear my choice](#)

Time left 0:4

In supervised Backpropagation learning algorithm, the change of the activation function w.r.t. the change of the net represent

- ☐ a. Training Set
- ☐ b. Weight
- ☒ c. Actual Outputs
- ☐ d. Sensitivity
- ☐ e. Targets
- ☐ f. Bias
- ☐ g. Forward Propagation

[Clear my choice](#)

Next

Question 19

Not yet answered

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Flag question

The set that has membership values of one and zero only is called:

- ☐ a. Universe of Discourse
- ☐ b. Fuzzy Set
- ☐ c. Dyadic Group
- ☒ d. Crisp Set
- ☐ e. Field
- ☐ f. Non-Classical Set
- ☐ g. Continuous Group

[Clear my choice](#)

Question 16

Not yet answered

Marked out of 3.00

Flag question

The following formula of (not Y or X) means:

- ☐ a. Non Fuzzy
- ☐ b. Linguistic
- ☐ c. Crisp
- ☐ d. not X entails Y
- ☒ e. Y entails X
- ☐ f. Deterministic
- ☐ g. Non-deterministic
- ☐ h. X entails Y

[Clear my choice](#)

Question 17

Not yet answered

Marked out of 3.00

Flag question

In Backpropagation algorithm, the following sequence of performed steps is valid:

- ☐ a. Backpropagation – Forward Propagation – Weight Update
- ☒ b. Forward Propagation – Backpropagation – Weight Update
- ☐ c. Backpropagation – Weight Update – Forward Propagation
- ☐ d. Weight Update – Forward Propagation – Backpropagation
- ☐ e. Weight Update – Backpropagation – Forward Propagation
- ☐ f. Forward Propagation – Weight Update – Backpropagation

[Clear my choice](#)

Question 15

Not yet answered

Marked out of 2.00

Flag question

The bias updating ($b^{new} = b^{old} + e$) is found in:

- ☐ a. Delta Learning Rule
- ☐ b. Smoothing Method
- ☐ c. Supervised Hebbian Learning
- ☐ d. Associative Learning
- ☐ e. Unsupervised Hebbian Learning
- ☐ f. Reinforced Learning
- ☒ g. Perceptron Learning

[Clear my choice](#)

Question 18

Not yet answered

Marked out of 4.00

Flag question

The square root of the linguistic value is called:

- ☐ a. Contrast
- ☐ b. Non-deterministic
- ☐ c. Non Fuzzy
- ☐ d. Stochastic
- ☒ e. Dilation
- ☐ f. Crisp
- ☐ g. Deterministic
- ☐ h. Chaotic

[Clear my choice](#)

Question 11

Not yet answered

Marked out of 3.00

Flag question

The universe of discourse $R = \{\text{Car, Water, Fire, Tree}\}$ represents:

- ☐ a. Discrete and Continuous
- ☒ b. Discrete and Non-Ordered
- ☐ c. Continuous and Ordered
- ☐ d. Ordered
- ☐ e. Discrete and Ordered
- ☐ f. Continuous
- ☐ g. Continuous and Non-Ordered

[Clear my choice](#)

Question 12

Not yet answered

Marked out of 2.00

Flag question

Linguistic values are used to form:

- ☐ a. Groups
- ☒ b. Fuzzy Partitions
- ☐ c. Universe of Discourse
- ☐ d. Crisp Sets
- ☐ e. Fuzzy Buffers
- ☐ f. Non Fuzzy Sets
- ☐ g. Classical Sets
- ☐ h. Fuzzy Delays

[Clear my choice](#)

Question 13

Not yet answered

Marked out of 3.00

Flag question

The many-sample performance index used in the Backpropagation algorithm is represented by:

- ☐ a. Actual Outputs
- ☐ b. Training Set
- ☐ c. Absolute Error
- ☐ d. Error Squared
- ☐ e. Target Outputs
- ☐ f. Error Only
- ☒ g. Mean Square Error

[Clear my choice](#)

Question 14

Not yet answered

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Flag question

The upper limit of the learning rate in LMS algorithm is:

- ☐ a. $2/3\lambda$
- ☐ b. 0.5
- ☒ c. $(1/2\lambda_{\max}) + (1/2\lambda_{\min})$
- ☐ d. 1.353
- ☐ e. 0.75
- ☐ f. $1/\lambda_{\min}$
- ☐ g. $1/(0.5\lambda_{\max} + \lambda_{\min})$
- ☐ h. $2.8(\lambda_{\min}/\lambda_{\max})$
- ☐ i. $3.7/\lambda_{\max}$

[Clear my choice](#)

2

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The upper limit for the learning rate in Widrow-Hoff supervised learning is:

- ☐ a. 1.5
- ☐ b. 0.5
- ☐ c. $2/3\lambda$
- ☒ d. $(1/2\lambda_{\max}) + (1/2\lambda_{\min})$
- ☐ e. $1/(0.5\lambda_{\max} + \lambda_{\min})$
- ☐ f. 0.75
- ☐ g. $1/\lambda_{\min}$
- ☐ h. $2.8(\lambda_{\min}/\lambda_{\max})$
- ☐ i. $3.7/\lambda_{\max}$

Question 10

Not yet answered

Marked out of 1.00

Flag question

In Perceptron learning algorithm, a single neuron can be used to classify:

- ☐ a. Five Patterns
- ☐ b. Quad Patterns
- ☐ c. Three Patterns
- ☒ d. Two Patterns
- ☐ e. Four Patterns
- ☐ f. One Pattern
- ☐ g. Seven Patterns

[Clear my choice](#)

Question 9

Not yet answered

Marked out of 4.00

Flag question

The variable Length in "Length is short" is called:

- ☐ a. Stochastic
- ☐ b. Non Fuzzy
- ☐ c. Chaotic
- ☐ d. Deterministic
- ☐ e. Non-Deterministic
- ☒ f. Linguistic
- ☐ g. Crisp
- ☐ h. Non-deterministic

[Clear my choice](#)

Question 2

Not yet answered

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Flag question

The activation function form $y=1/(1+e^{-n})$ represents:

- ☐ a. Hyperbolic Tangent Sigmoid
- ☐ b. Saturating Linear
- ☒ c. Log-Sigmoid Squared
- ☐ d. Logistic
- ☐ e. Linear
- ☐ f. Quadratic
- ☐ g. Cubic Hyperbolic
- ☐ h. Hard Limit

[Clear my choice](#)

Question 7

Not yet answered

Marked out of 3.00

Flag question

The points at which the value of the membership function is 50% of its maximum are called:

- ☒ a. Crossover
- ☐ b. Continuous and Non-Ordered
- ☐ c. Continuous and Ordered
- ☐ d. Ordered
- ☐ e. Passing points
- ☐ f. Continuous
- ☐ g. Discrete and Non-Ordered
- ☐ h. Core

[Clear my choice](#)

Question 8

Not yet answered

Marked out of 2.00

Flag question

In the Backpropagation algorithm, the array that contains the activation functions derivatives is called:

- ☐ a. Actual Array
- ☐ b. Target Array
- ☐ c. Auto Correlation
- ☒ d. Jacobian
- ☐ e. Hessian
- ☐ f. Input Vector
- ☐ g. Correlation

[Clear my choice](#)

Question 5

Not yet answered

Marked out of 2.00

Flag question

In LMS Algorithm, the expression $[R]^{-1} [h]$ represents:

- ☐ a. Repeated Minima
- ☐ b. Local Maximum
- ☐ c. Repeated Maxima
- ☐ d. Global Minimum
- ☐ e. Local and Global Maximum
- ☒ f. Global Maximum
- ☐ g. Local Minimum

[Clear my choice](#)

Question 6

Not yet answered

Marked out of 4.00

Flag question

The S-norm requirement of $S(x, y) = S(y, x)$ is called:

- ☐ a. Distributivity
- ☒ b. Commutativity
- ☐ c. Boundary
- ☐ d. Ordered
- ☐ e. Crossover
- ☐ f. Continuous
- ☐ g. Continuous and Non-Ordered
- ☐ h. Discrete and Ordered

[Clear my choice](#)

Question 2

Not yet answered

Marked out of 1.00

Flag question

The updating $W^{new} = W^{old} + \alpha a_q p_q^T$ resembles:

- ☐ a. Smoothing Learning
- ☐ b. Reinforced Learning
- ☐ c. LMS Algorithm
- ☒ d. Unsupervised Hebbian
- ☐ e. Unsupervised Reinforced
- ☐ f. Backpropagation
- ☐ g. Delta Rule
- ☐ h. Widrow-Hoff Learning
- ☐ i. Perceptron Method

[Clear my choice](#)

Question 1

Not yet answered

Marked out of 3.00

Flag question

For noise elimination system, the basic element utilized within such supervised learning design is:

- ☐ a. Noise Source
- ☐ b. Fixed-Valued Filter
- ☐ c. RC Filter
- ☐ d. RLC Filter
- ☒ e. Adaptive Filter
- ☐ f. Non-Adjustable Filter
- ☐ g. RL Filter
- ☐ h. Noise Path Filter

[Clear my choice](#)

For noise cancellation system which was studied as an application of supervised learning, the basic element utilized within such design for supervised learning is:

- ☐ a. RLC Filter
- ☐ b. RCL Filter
- ☐ c. Non-Adjustable Filter
- ☐ d. Fixed-Valued Filter
- ☐ e. RC Filter
- ☐ f. Noise Path Filter
- ☒ g. Adaptive Filter
- ☐ h. Noise Source

[Clear my choice](#)

4/37

A fuzzy set is convex if:

- ☐ a. One alpha cut is convex
- ☐ b. It is Linguistic
- ☐ c. Beta is convex
- ☐ d. Non Fuzzy
- ☐ e. It is Crisp
- ☐ f. All alpha cuts are nonconvex
- ☐ g. It is Deterministic
- ☐ h. at least half of the alpha cuts are nonconvex
- ☒ i. All alpha cuts are convex
- ☐ j. not X entails Y
- ☐ k. It is Non-deterministic

A Fuzzy relation is:

- ☐ a. not Y entails not X
- ☐ b. Three-Dimensional μ
- ☐ c. not X entails not Y
- ☒ d. Two-Dimensional μ
- ☐ e. Crisp
- ☐ f. Pseudo-Crisp
- ☐ g. Non Fuzzy
- ☐ h. Deterministic

[Clear my choice](#)

Time left

If input fuzzy variable #1 has four linguistic values, input fuzzy variable #2 has three linguistic values, and the output fuzzy variable has five linguistic values, then the total number of rules is

- ☐ a. 13
- ☐ b. $3/4$
- ☐ c. 8
- ☐ d. 11
- ☐ e. 16
- ☐ f. 2
- ☐ g. 15
- ☐ h. $4/3$
- ☐ i. 21
- ☐ j. 5
- ☐ k. 10
- ☐ l. 7
- ☐ m. 3
- ☐ n. 12

Search

Question 2

Not yet answered
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Flag question

One approach for defuzzification is to use:

- ☐ a. Min-Max Decomposition
- ☐ b. Max-Min Decomposition
- ☒ c. not Y entails not X
- ☐ d. x AND y
- ☐ e. Max-Min Composition
- ☐ f. Min-Max Composition
- ☐ g. Center of Gravitation
- ☐ h. not X entails not Y

Clear my choice

7/37

For the function $(U + V') \cdot (U' + V)$, pattern classification can be done using:

- ☐ a. Cannot be Classified
- ☐ b. One Neuron
- ☐ c. Single Decision Boundary
- ☐ d. One Perceptron-Based Neuron
- ☒ e. Multilayer Perceptron
- ☐ f. One Classification Boundary

[Clear my choice](#)

12/37

Time left

In Backpropagation algorithm, the matrix that contains the transfer functions derivatives is called:

- ☐ a. Auto Correlation Matrix
- ☐ b. Hessian Matrix
- ☐ c. Input Correlation Matrix
- ☐ d. Correlation Matrix
- ☒ e. Jacobian Matrix
- ☐ f. Target Matrix
- ☐ g. Input Vector Matrix

[Clear my choice](#)

14/37

5
of

A Fuzzy relation is:

- ☐ a. not Y entails not X
- ☐ b. not X entails not Y
- ☐ c. Crisp
- ☐ d. Deterministic
- ☐ e. Non Fuzzy
- ☐ f. Pseudo-Crisp
- ☒ g. Two-Dimensional μ
- ☐ h. Three-Dimensional μ

Fuzzy Yager and Sugeno complements are the same when the v respectively:

- ☐ a. (0.5, -0.5)
- ☐ b. (20, 1.5)
- ☐ c. (0.75, 1.0)
- ☐ d. (2/3, 3/2)
- ☐ e. (0.5, 0.5)
- ☐ f. (2, 0.7)
- ☐ g. (3/2, 2/3)
- ☐ h. (0, 1)
- ☐ i. (1, 0)
- ☐ j. (-0.5, 0.4)
- ☐ k. (-0.5, 0.5)

if it was (s, w) respectively
=> answer is (0,1)

if it was (w, s) respectively
=> answer is (1,0)

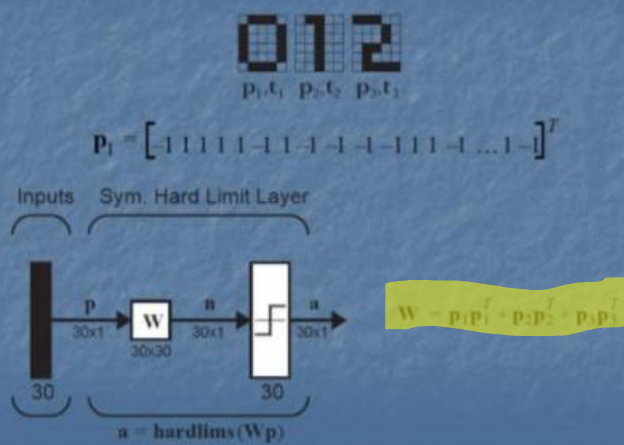
■ Sugeno's complement:

$$N_s(a) = \frac{1-a}{1+sa}$$

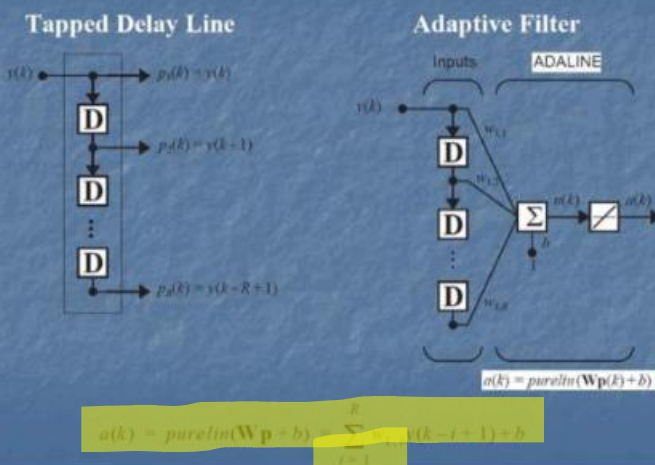
■ Yager's complement:

$$N_w(a) = (1-a^w)^{1/w}$$

Autoassociative Memory



Adaptive Filtering



Fuzzy Union: T-conorm or S-norm

- Basic requirements:
 - Boundary: $S(1, 1) = 1$, $S(a, 0) = S(0, a) = a$
 - Monotonicity: $S(a, b) < S(c, d)$ if $a < c$ and $b < d$
 - Commutativity: $S(a, b) = S(b, a)$
 - Associativity: $S(a, S(b, c)) = S(S(a, b), c)$
- Four examples:
 - Maximum: $S_m(a, b)$
 - Algebraic sum: $S_a(a, b)$
 - Bounded sum: $S_b(a, b)$
 - Drastic sum: $S_d(a, b)$

-MATRIX A Represent: Hessian matrix

-MATRIX R Represent: correlation matrix

Stationary Point

Hessian Matrix:

$$\mathbf{A} = 2\mathbf{R}$$

The correlation matrix **R** must be at least positive semidefinite. If there are any zero eigenvalues, the performance index will either have a weak minimum or else no stationary point, otherwise there will be a unique **global minimum** $\mathbf{x}^*$.

$$\nabla F(\mathbf{x}) = \nabla \left(c + \mathbf{d}^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} \right) = \mathbf{d} + \mathbf{A} \mathbf{x} = -2\mathbf{h} + 2\mathbf{R} \mathbf{x}$$
$$-2\mathbf{h} + 2\mathbf{R} \mathbf{x} = 0$$

If **R** is positive definite:

$$\mathbf{x}^* = \mathbf{R}^{-1} \mathbf{h}$$

-Multi neuron perceptron can classify input vector :

2^s categories

Multiple-Neuron Perceptron

Each neuron will have its own decision boundary

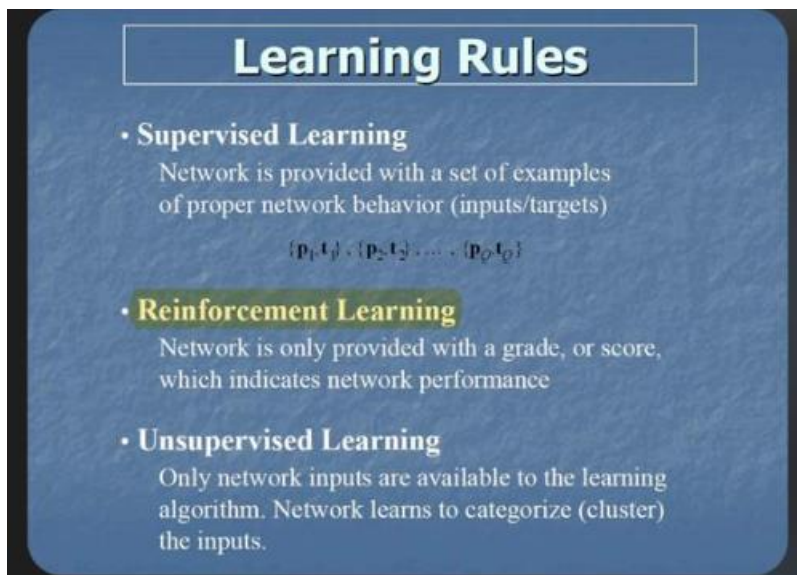
$$\mathbf{w}_i^T \mathbf{p} + b_i = 0$$

A single neuron can classify input vectors into two categories.

A multi-neuron perceptron can classify input vectors into **2^s** categories.

-Network only provided with grader score :

Reinforcement



Learning Rules

- **Supervised Learning**
Network is provided with a set of examples of proper network behavior (inputs/targets)
$$\{p_1, t_1\}, \{p_2, t_2\}, \dots, \{p_P, t_P\}$$
- **Reinforcement Learning**
Network is only provided with a grade, or score, which indicates network performance
- **Unsupervised Learning**
Only network inputs are available to the learning algorithm. Network learns to categorize (cluster) the inputs.

-Number of the neuron in the network (Hopfield net work):

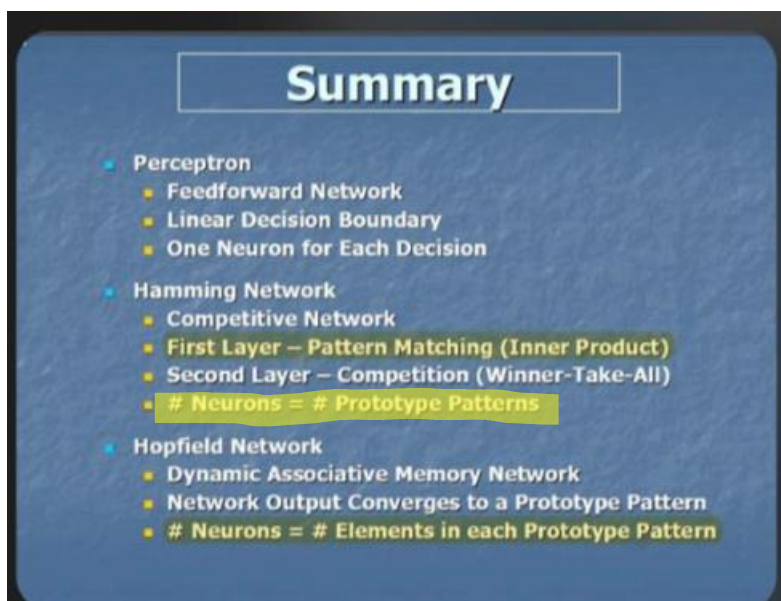
no of element of each prototype pattern

-In Hamming Network the first layer :

first layer -pattern matching (Inner product)

-In hamming network the number of neurons :

The no. of prototype pattern



Summary

- **Perceptron**
 - Feedforward Network
 - Linear Decision Boundary
 - One Neuron for Each Decision
- **Hamming Network**
 - Competitive Network
 - **First Layer – Pattern Matching (Inner Product)**
 - **Second Layer – Competition (Winner-Take-All)**
 - **# Neurons = # Prototype Patterns**
- **Hopfield Network**
 - Dynamic Associative Memory Network
 - Network Output Converges to a Prototype Pattern
 - **# Neurons = # Elements in each Prototype Pattern**

$$* W = a_2 f_2^T \rightarrow \text{Non-linear Approximation}$$

-only network inputs are available to the learning algorithm network learns to categorize (cluster) the input :

Unsupervised

-Unsupervised used to :

categorize (cluster) the input

MF Formulation

■ **Triangular MF:**
$$\text{trimf} (x; a, b, c) = \max \left(\min \left(\frac{x-a}{b-a}, \frac{c-x}{c-b} \right), 0 \right)$$

Trapezoidal MF:
$$\text{trapmf} (x; a, b, c, d) = \max \left(\min \left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c} \right), 0 \right)$$

Gaussian MF:
$$\text{gaussmf} (x; a, b, c) = e^{-\frac{1}{2} \left(\frac{x-c}{\sigma} \right)^2}$$

Generalized bell MF:
$$\text{gbellmf} (x; a, b, c) = \frac{1}{1 + \left| \frac{x-c}{b} \right|^{2b}}$$

Unanswered questions

-what is the difference btw the classical fuzzy and logical fuzzy :